Zhiyuan Ouyang

2nd-year Ph.D. @ East China Normal University / Shanghai Institute of Intelligence

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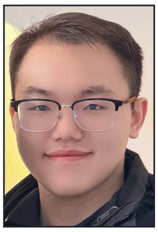

Summary

My research centers on **Generative Models**, focusing on Diffusion / Flow-Matching architectures and underlying topologies to enhance generation efficiency and controllability. I possess **full-stack edge AI delivery capabilities**, independently handling core algorithm design, model training, and native edge deployment. In AI engineering, I have **large-scale distributed training** expertise, familiar with Slurm and multi-node DDP cluster scheduling, with hands-on experience in concurrent training and communication tuning. Furthermore, I specialize in **custom inference infrastructure**, excelling in inference acceleration and extreme VRAM optimization using TensorRT and ONNX Runtime. I am also interested in **blockchain trading systems** and on-chain execution infrastructure, with end-to-end experience in designing and delivering strategy engines, risk-controlled execution, and stateful trading pipelines.

Education

ECNU/SII 2nd-year Ph.D. in Computational Mathematics — Shanghai, China

Advisor: Prof. Junchi Yan

Honors & Awards: National Scholarship for PhD Students, China (Top 1.5%)

Competition Awards: C/C++ Programming Contest, Collegiate Division (Provincial First Prize, Top 5%)

Research Interests: Foundational Generative Models / Multi-modal Generation / Controllable Generation / Time-series Analysis / On-chain Execution Systems

Publications

Zhiyuan Ouyang (First Author). Primal-Spectral Generative Modeling: Fast Analytical Generation via Pseudoinverse Lévy Inversion. *Published in **ICML 2026** (CCF-A)*

Zhiyuan Ouyang (First Author). A Novel Network for Resolving Subjective Masking Differences and Accurate Thyroid Nodule Diagnosis. *Published in **Computational Biology and Chemistry** (JCR-Q2)*

Zhiyuan Ouyang (First Author). Model Detection for Grey Forecasting Model with Polynomial Term. *Published in **Communications in Statistics** (JCR-Q3)*

Projects

MimicPolymarket On-chain Copy Trading Engine (Independent Developer) **Repo** **Live Account** Feb 2026 – Present

- **Core Architecture:** Designed an industrial-grade execution engine with multi-address matrix concurrency, compatible with EOA, Safe, and Deposit Wallet modes. Deeply integrated Polymarket's Relayer routing and the **POLY_1271** signature protocol, achieving a **352% monthly return** on live accounts.
- **Risk Management & Execution:** Built a configurable risk management matrix per trader (mimic ratio, position limits, slippage, etc.). Employed a conservative execution logic of balance protection combined with Limit Orders for opening positions, and executed FOK and dynamic position reduction based on the order book's Best Bid for closing, effectively reducing Market Impact and abnormal trading risks.
- **Signal Aggregation & Fault Tolerance:** Developed a fixed-time-window **Order Aggregation** engine to merge high-frequency, fragmented Snipe/MEV signals and generate synthetic trades. Persisted full-lifecycle states to **MongoDB** in real-time, achieving a low-latency, highly available, and auditable end-to-end on-chain trading loop via serial queues and state write-back mechanisms.

Generative Modeling in Characteristic Function Space (Outstanding Ph.D. Project, ICML 2026) **Paper**
Nov 2025 – May 2026

- **Theoretical Innovation:** Explored the evolution mechanism from Primal Data Distribution to Characteristic Function. Transformed the distribution into a continuous function via Spectral Methods, solving the challenge of neural networks reconstructing global topology from sparse samples, and providing theoretical guarantees for convergence.
- **Algorithm Architecture:** Proposed the **PriSpecNet** architecture and designed a single-step evaluation (1-NFE) Pseudoinverse Lévy Inversion (PiLI) solver. This transforms the generation process into a Closed-form Analytic prob-

lem, completely eliminating the reliance on iterative Integration while maintaining full compatibility with Stochastic Interpolation.

- **Application Performance:** Unified generation and forecasting in the time series domain, reducing Context-FID metrics on Sines, Solar, ETTh, and Stock datasets by 50.0%, 41.5%, 80.6%, and 63.1%, respectively. In the ImageNet 256×256 generation task, achieved an FID of 1.66 with only **26 Gflops** computational overhead, realizing a **170x** computation reduction compared to 25-NFE DPM-Solver++.

Controllable Video Generation Project (University-Industry Collaboration, Project Lead) Oct 2025 – Jun 2026

- **Project Leadership:** Served as project lead coordinating university and industry requirements, driving requirement decomposition, technical decisions, and milestone delivery; investigated representative approaches including **Control-A-Video** and **ControlVideo**, defining a **Video ControlNet**-centric solution.
- **Generation Pipeline Construction:** Led the development of a controllable video generation workflow based on **ComfyUI** and **Stable Diffusion**, ensuring the stability and scalability of the end-to-end generation pipeline.
- **Data Toolchain Development:** Developed an automated motion trajectory annotation tool based on the **Sapiens** vision model. Achieved high-precision human keypoint extraction and temporal skeleton tracking across video frames, building high-quality control signal datasets that significantly improved the model's ability to follow complex actions.

AIGC Customized Models & Short Video Pipeline (Commercial Project, Project Lead) Sep 2023 – Sep 2024

- **Technical Consulting & Commercial Landing:** Provided end-to-end AIGC technical consulting services for e-commerce and home furnishing enterprise clients. Analyzed pain points in business content production and formulated generative AI solutions, driving the commercial deployment of AI technologies in real-world production environments.
- **Enterprise-Grade Model Customization & Deployment:** Established a high-standard data processing pipeline for industry-specific private data. Utilized the **Stable Diffusion** architecture and **LoRA** fine-tuning techniques to train highly customized vertical models, overcoming challenges in the high-fidelity restoration of specific product features and complex decor styles, and executed enterprise-grade on-premise deployment.
- **Industrial Generation Pipeline Delivery:** Developed and delivered an automated, one-click short video generation Workflow driven by text and image assets. Integrated core modules such as model inference, temporal control, and video synthesis, encapsulating them into out-of-the-box enterprise services via standard APIs, significantly reducing clients' video creation barriers and operational costs.

Internship

Baizhi Xunlian Technology Co., Ltd. (Head of AI Department) Sep 2023 – Sep 2024

- **Generative Model R&D:** Led the construction of visual generation Pipelines based on **ComfyUI** and **Stable Diffusion**. Introduced SFT and human preference alignment (RL) for vertical domain Fine-tuning, delivering highly controllable, End-to-End customized image generation solutions.
- **AI Engineering & Service Deployment:** Led AIGC infrastructure development and Model-as-a-Service (MaaS) deployment. Built the complete pipeline from model packaging and API encapsulation to high-concurrency cloud inference; designed GPU resource scheduling and task queue mechanisms to ensure high availability for generative tasks.
- **Team Management:** Comprehensively managed the technical evolution and Agile R&D of the AI department. Provided AIGC commercial landing consulting; established internal engineering standards, led technical training, and drove efficient cross-functional project delivery.

Technical Skills

Programming Languages: Python - proficient in algorithm modeling and data analysis; C++ / TypeScript - utilized for high-performance inference and production-grade engineering execution; adept at using AI coding assistants.

Algorithm & Training: Deep understanding of the underlying principles of Diffusion / Flow-Matching vision generative models; proficient in utilizing SFT and LoRA for commercial deployment and landing; skilled in PyTorch, JAX, and distributed training.

Engineering & Architecture: Experienced in AIGC infrastructure and Model-as-a-Service (MaaS) deployment; proficient in GPU resource scheduling, API encapsulation, and workflow (e.g., ComfyUI) development.

Data & Toolchain: Proficient in large-scale multimodal data synthesis; skilled in utilizing vision models (e.g., Sapiens) to build automated annotation toolchains for high-quality control datasets.